ROAD-04-IP-STRATEGY

ROAD-04: INTELLECTUAL PROPERTY STRATEGY

Carrington Storm Motors — Project AEGIS

Patent Landscape, Trade Secrets, Defensive Publishing & Freedom-to-Operate

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EXECUTIVE SUMMARY

Carrington Storm Motors sits at the intersection of six distinct patent landscapes: EMI shielding materials, MXene chemistry, GIC-hardened electronics, dielectric building materials, amphibious vessel design, and proton-based communication. The CSM IP portfolio — currently at the invention disclosure stage with zero filed patents — represents a first-mover opportunity in the emerging "electromagnetic resilience" market category. This document maps the patent landscape, identifies 42 patentable inventions across the CSM portfolio, defines the trade secret vs. patent boundary, outlines a defensive publishing strategy, assesses freedom-to-operate for all major product lines, and establishes a licensing strategy for the post-patent commercial phase.

The IP Strategy in One Sentence: Patent the composition-of-matter and device claims that competitors would reverse-engineer; protect as trade secrets the manufacturing processes that competitors cannot reproduce without CSM's proprietary know-how; defensively publish around competitor patent thickets to preserve freedom-to-operate; and license the AEGIS CERTIFIED mark to create a quality brand premium.

SECTION 1: PATENT LANDSCAPE ANALYSIS

1.1 EMI Shielding Materials (IPC H05K 9/00, C09D 5/24)

Competitive Landscape: - **Laird Technologies / DuPont:** Dominant in metal-based EMI shielding (copper mesh, nickel paints, conductive elastomers). Patent portfolio focused on reflection-dominant shielding with metallic fillers. Date of earliest priority: 1998. Expiration of key patents: 2023–2028. - **Parker Hannifin (Chomerics):** Conductive coatings and gaskets. Patent US10,524,382 on conductive fabric gaskets (expires 2036). Relevant but does not cover absorption-dominant non-metallic shielding. - **Henkel / Bergquist:** Thermal + EMI gap filler pads. Patent family on boron nitride + conductive filler composites. Expiration: 2030–2035.

CSM White Space: No existing patent covers (a) MXene $Ti_3C_2T_x$ as the primary EMI shielding agent in an architectural or vehicle coating, (b) discontinuous tile pattern achieving DC isolation while maintaining RF continuity via capacitive coupling between tiles, (c) absorption-dominant multilayer ceramic EMI shield with >148 dB SE across 1 kHz–10 GHz, or (d) the complete 7-layer Aegis-C stack architecture as an integrated system.

Freedom-to-Operate Assessment: CSM products do not infringe any existing EMI shielding patents because (1) we use absorption-dominant MXene, not reflection-dominant metals; (2) our discontinuous tile architecture is unique; (3) our multilayer stack integrates materials (ZrB_2-SiC, YInMn Blue, ZrO_2 foam, FSS) that have never been combined in an EMI shielding context. **FTO Rating: GREEN — no blocking patents identified.**

1.2 MXene Chemistry and Processing (IPC C01B 32/00, C01G)

Competitive Landscape: - **Drexel University (Yury Gogotsi / Michel Barsoum):** Foundational MXene patents. US9,193,595 — "Two-dimensional titanium carbide and methods of making the same" (filed 2012, expires 2032). Covers $Ti_3C_2T_x$ composition and HF etching method. This is the dominant blocking patent in the

MXene space. - **Drexel University:** US10,947,112 — covers MXene-polymer composites (expires 2038). - **Drexel University:** US11,021,374 — covers MXene EMI shielding (expires 2039). Claim 1: "An electromagnetic interference shielding article comprising a substrate and a coating comprising $Ti_3C_2T_x$ MXene." THIS IS DIRECTLY RELEVANT to CSM's core MXene coating application.

FTO Analysis — Drexel MXene Patents: CSM's current MXene usage falls within the scope of Drexel's patent claims (US9,193,595 composition of matter; US11,021,374 EMI shielding article). **Resolution Path:** 1. **License negotiation with Drexel:** Negotiate an exclusive or non-exclusive license to the Drexel MXene patent portfolio. Estimated royalty: 2–4% of net sales on MXene-containing products. Drexel has expressed interest in commercial partnerships. Aegis MXene Lab (hypothetical CSM spin-in) could serve as the licensing entity. 2. **Invent around:** Develop an alternative fluoride-free MXene synthesis method that produces MXene with different surface termination chemistry (e.g., all-oxygen termination vs. OH/F mixed termination). If the resulting material has measurably different properties (e.g., higher conductivity, different interlayer spacing), it may fall outside the literal scope of Drexel's composition-of-matter claims. **This is the preferred CSM strategy and is already underway per the fluoride-free electrochemical etching development program.** 3. **Patent non-assertion agreement:** For public-benefit applications (emergency communications, humanitarian disaster response), negotiate a royalty-free license from Drexel under their social-impact licensing program.

FTO Rating: AMBER — licensing or invent-around required for MXene composition-of-matter. CSM's process innovations (fluoride-free etching, spray coating at 45 μm with <2 μm tolerance, discontinuous tile patterning) are independently patentable and do not depend on Drexel's process claims.

1.3 Dielectric Building Materials (IPC E04B 1/92, E04C 2/00)

Competitive Landscape: - **Sika AG, BASF:** Dominant in construction chemicals and composite reinforcement. No patent coverage for EMI-shielded building panels. - **Kingspan, Metl-Span:** Insulated metal panels (IMP) for building envelopes. Metal-based — completely opposite to CSM's dielectric approach. - **Owens Corning:** Fiberglass composite rebar (GFRP). Patents on rebar composition, but no EMI function claimed.

CSM White Space: No patent covers (a) a residential building wall panel designed for GIC protection, (b) the integration of BFRP rebar with geopolymer concrete for EMI-resistant foundations, (c) a complete non-metallic building system (roof, walls, foundation, wiring, plumbing, HVAC) designed for Carrington-level electromagnetic survivability. **FTO Rating: GREEN — no blocking patents.**

1.4 Amphibious Vessel Design (IPC B63B 1/00, B60F 3/00)

Competitive Landscape: - **Gibbs Technologies:** Amphibious vehicle patents (US7,255,047, expires 2024; wheel-based amphibious, not hexapedal). - **DARPA / Boston Dynamics:** Legged robot patents (US8,972,052 — BigDog; covers hydraulic actuation for legged locomotion, expires 2032). - **US Navy:** Amphibious assault vehicle patents (AAV-7, wheel/track-based, not walking).

CSM White Space: No patent covers (a) a hexapedal walking mechanism for a vessel hull, (b) Archimedes screw propulsion for a ship-scale vessel (vs. drone-scale, which has prior art), (c) an amphibious transition mechanism using smart rope actuators (see CSMFAB0113). **FTO Rating: GREEN — CSM's amphibious concepts are novel relative to all existing art.**

1.5 Proton-Based Communication (IPC H01L 29/66, H04B 13/00)

Competitive Landscape: - **Nafion-based H-FET research:** Literature only (academic papers from Harvard, MIT, Stanford). No patents identified on complete protonic logic gate families or protonic communication systems. - **Molecular communication (IEEE P1906.1):** Standard for nanoscale molecular communication. Not applicable to macroscale protonic radio. - **Underwater acoustic communication:** Alternative non-EM communication method. Not GIC-immune (piezoelectric transducers still susceptible).

CSM White Space: This is a GREEN FIELD. No existing patents on (a) protonic logic gates for EMI-immune computing/communication, (b) a complete protonic communication network architecture, (c) Pd-free H-FET using MXene gate electrodes, (d) hybrid protonic-photonic communication protocol stack. **FTO Rating: GREEN — pioneering invention with zero blocking patents.**

1.6 Piezoelectric Actuators and Energy Harvesting (IPC H02N 2/00)

Competitive Landscape: - **Physik Instrumente (PI):** Dominant in precision piezoelectric actuators. Patent portfolio on stacked piezo actuators, ultrasonic motors. Relevant to Phantom actuator design. - **CTS Corporation:** $KNbO_3$-based piezoelectrics. CSM's $KNbO_3$-$BaTiO_3$ is a different composition (barium titanate addition changes Curie temperature and d_{33}). - **MicroGen Systems:** Piezoelectric energy harvesting for IoT. Patents on cantilever-based MEMS harvesters — not applicable to CSM's bulk piezo transducers.

FTO Rating: GREEN — CSM's $KNbO_3$-$BaTiO_3$ composition is distinct from existing patented formulations. Actuator configurations (32-element Phantom joint, bone conduction transducer) are novel applications.

SECTION 2: CSM PATENTABLE INVENTIONS — 42 CLAIMS IDENTIFIED

Category A: Composition of Matter (Highest IP Value — File First)

ID	Invention	Key Claims	Priority Date Target	Lead Inventor
A1	Fluoride-free electrochemical MXene synthesis	Method of producing $Ti_3C_2T_x$ MXene via electrochemical etching in fluoride-free electrolyte, producing MXene with <5 at% fluorine surface termination and >60 at% oxygen termination	Q4 2026	Nash
A2	MXene-polymer composite EMI shield with discontinuous tile architecture	An EMI shielding article comprising MXene tiles of dimension 1–5 cm separated by insulating gaps of 10–100 µm, achieving >90 dB SE at 1 kHz–10 GHz with DC resistance between tiles $>10^8$ Ω	Q4 2026	Nash
A3	$KNbO_3$-$BaTiO_3$ lead-free piezoelectric composition with enhanced Curie temperature	A piezoelectric ceramic comprising $(1-x)KNbO_3$-$xBaTiO_3$ where $0.05 \leq x \leq 0.15$, exhibiting $d_{33}>250$ pC/N and $T_c>300$°C	Q1 2027	Kade
A4	YInMn Blue pigment with $CsPbBr_3$ perovskite quantum dot enhancement	A near-infrared reflective pigment comprising $YIn_{1-x}Mn_xO_3$ matrix with $CsPbBr_3$ quantum dots embedded for UV down-conversion and enhanced chromatic saturation	Q1 2027	Voss
A5	$BaZrO_3$-yttria solid-state proton conductor for H-FET applications	A proton-conducting ceramic comprising $BaZr_{1-x}Y_xO_{3-\delta}$ where $0.10 \leq x \leq 0.25$, exhibiting proton conductivity >0.01	Q3 2027	Vance

S/cm at 400°C

Category B: Device & System Claims

ID	Invention	Key Claims	Priority Date	Lead Inventor
B1	Dielectric Citadel Protocol — 7-layer electromagnetic shielding panel	A building panel comprising (from exterior to interior): UHTC outer layer, MXene EMI shield, ceramic foam core, pigment layer, frequency-selective surface, ceramic bracket layer, and compression gasket joint system	Q4 2026	Steele
B2	Phantom MK-1 — GIC-hardened humanoid robot architecture	An autonomous humanoid robot wherein all structural, actuator, sensor, and power system components are non-ferromagnetic and have electrical resistance to chassis >10^8 Ω	Q1 2027	Calder
B3	Charlemagne-class vessel electromagnetic hardening system	A marine vessel electromagnetic protection system comprising: MXene hull coating, BFRP structural elements, PMMA POF data network, and $LiFePO_4$ battery system, achieving <0.03% hull-induced current relative to unprotected baseline	Q2 2027	Vaun
B4	Protonic communication transceiver (H-FET based)	A communication device comprising: an H-FET logic gate family fabricated on Nafion membrane with MXene gate electrodes, a PMMA POF optical interface, and a $KNbO_3$-$BaTiO_3$ piezoelectric power source	Q3 2027	Vance
B5	Amphibious vessel hexapedal walking mechanism	A marine vessel hull comprising six articulated legs, each leg including smart rope actuators (Archimedes muscle design), enabling transition from buoyant to terrestrial locomotion	Q4 2027	Vaun

B6	Smart Rope Actuator (Archimedes Muscle)	An actuator comprising a helically wound fiber bundle with embedded dielectric tensioning elements, achieving linear contraction via rotary input without metallic components	Q2 2027	Fen
B7	Neural-Grip active MRE steering wheel	A vehicle steering wheel comprising: a magnetorheological elastomer layer with embedded piezoelectric elements, achieving >90% vibration cancellation at motor torque ripple frequency while capacitively coupling Schumann Resonance to driver palm	Q1 2027	Nash
B8	Cervical-Guard viscoelastic headrest with bone conduction entrainment	An automotive headrest comprising: active vibration damping at cranium resonance frequency, and a bone conduction transducer array delivering 7.83 Hz Schumann entrainment signal to temporal bone	Q1 2027	Kade
B9	Mag-Float diamagnetic seat rail system	A vehicle seat adjustment mechanism using pyrolytic graphite diamagnetic levitation with permanent magnet track, eliminating all metallic sliding contact	Q2 2027	Draven
B10	Silence-Block LRAM door insert	An automotive door cavity insert comprising a locally resonant acoustic metamaterial array tuned to 82 Hz, achieving >18 dB transmission loss at panel coincidence frequency	Q1 2027	Cross
B11	Thorax-Calm shear-thickening fluid seatbelt	A seatbelt restraint system incorporating shear-thickening fluid in a flexible bladder along the belt path, increasing energy absorption by >100%	Q2 2027	Arden

		during rapid deceleration		
B12	Stellar-Tint VO_2 electrochromic window film (4-mode)	A vehicle window film comprising a VO_2-based electrochromic layer capable of switching between 4 optical states: NIR-reflective, emergency beacon (amber), calming (green 534 nm), and adaptive camouflage (NIR scatter)	Q1 2027	Voss
B13	Fluid-Damp MRF-based gear shift interface	A vehicle shift mechanism using magnetorheological fluid to provide variable haptic resistance, eliminating metallic detent mechanisms	Q2 2027	Fen
B14	Ulnar-Rest constrained-layer damping armrest	A vehicle armrest comprising a constrained-layer damping structure with loss factor $\eta>0.30$, reducing 20–50 Hz vibration transmission to occupant forearm	Q2 2027	Dorne
B15	Pet-Safe infrasound cargo liner	A vehicle cargo area liner integrating infrasound-absorbing metamaterial and MXene EMI shielding, reducing animal stress during transport	Q1 2027	Solara
B16	Cloud-Nest aerogel child seat	A child safety seat comprising a non-metallic aerogel composite shell with electrocaloric active cooling and integrated EMI shielding	Q1 2027	Rook
B17	Way-Finder haptic insole navigation system	A footwear insole with embedded piezoelectric transducers providing haptic directional cues for pedestrian navigation without visual or auditory dependency	Q1 2027	Solven
B18	Aegis Embark grounding-sole footwear	Footwear comprising a conductive CNT-polymer sole with controlled surface resistivity (10^6–10^8 Ω/sq) for safe body	Q2 2027	Kind

		charge dissipation while maintaining GIC isolation		
B19	Aegis Home complete non-metallic building system	A residential building wherein the foundation (geopolymer concrete with BFRP rebar), walls (Aegis-C panels), roof, wiring (PMMA POF), plumbing (GFRP), HVAC, and all fixtures are non-metallic and achieve >94% GIC reduction	Q3 2027	Nyx
B20	Phoenix Protocol — MXene indium recovery from end-of-life panels	A method for recovering indium from MXene-containing composite panels via D2EHPA solvent extraction, achieving >94% recovery efficiency	Q4 2027	Steele

Category C: Software & Methods

ID	Invention	Key Claims	Priority Date	Lead Inventor
C1	LandOLil — GIC risk assessment and hardening verification platform	A system for continuous monitoring of geomagnetic induced current risk using distributed sensors, dielectric installation logging, and automated premium credit calculation for insurance carriers	Q3 2026	Zirconia
C2	Marmalade — Offline-first emergency preparedness application	A mobile application with offline-first architecture for managing hardened residential systems, including inventory tracking, mesh communication, and scenario simulation	Q3 2026	Zirconia
C3	CSM Digital Twin — electromagnetic resilience simulation platform	A digital twin system integrating CST EM simulation, real-time GIC sensor data, and physics-based degradation modeling for predicting infrastructure vulnerability to Carrington-class events	Q4 2027	Cross
		A training system for civilian emergency		

C4	“What If” Scenario Simulator	preparedness using Monte Carlo simulation of Carrington scenarios with branching decision trees and consequence modeling	Q1 2028	Zirconia
C5	Protonic network mesh protocol (LoRa + POF hybrid)	A communication protocol for hybrid protonic-photonic mesh networks with self-healing topology and priority-based message queuing for post-disaster environments	Q4 2027	Vance

Category D: Manufacturing Processes

ID	Invention	Key Claims	Priority Date	Lead Inventor
D1	Flash sintering for ZrB_2-SiC UHTC ceramics (38% energy reduction)	Method of sintering ZrB_2-SiC composites by applying DC electric field during heating, achieving >98% density at 300°C below conventional sintering temperature	Q1 2027	Nash
D2	MXene spray coating uniformity control (45±2 μm)	Method for depositing MXene thin films with <5% thickness variation over areas >1 m² via multi-nozzle spray system with real-time spectroscopic feedback	Q2 2027	Nash
D3	BFRP/Elium pultrusion with in-line ultrasonic inspection	Continuous pultrusion method with in-line ultrasonic C-scan for void detection, enabling 100% inspection without slowing production	Q1 2027	Cross
D4	Discontinuous MXene tile coating via stencil spray	Method for creating DC-isolated MXene tiles using sacrificial stencil during spray coating, enabling capacitive RF coupling between tiles while maintaining $>10^8$ Ω DC resistance	Q4 2026	Nash
D5	H-FET logic gate fabrication on flexible	Method for fabricating AND, OR, NOT, NAND, NOR logic gates via series/parallel Nafion	Q3 2027	Vance

	Nafion substrate	membrane channel arrangements with MXene gate electrodes, achieving $>10^5$ on/off ratio		
D6	Aegis-C panel automated assembly (LOM + FSS photolithography)	Integrated manufacturing line combining Laminated Object Manufacturing tape casting with in-line frequency-selective surface photolithographic etching for continuous panel production	Q2 2027	Steele

Category E: Test & Measurement Methods

ID	Invention	Key Claims	Priority Date	Lead Inventor
E1	5-Phase Phantom EM Qualification Protocol	Test methodology comprising sequential EM exposure, vibration, thermal cycling, combined environment, and degraded-mode operation for certifying autonomous systems for post-Carrington deployment	Q3 2026	Zirconia
E2	In-situ GIC monitoring sensor network	Distributed sensor system for measuring ground-induced currents in residential and infrastructure settings with LoRa mesh data backhaul	Q1 2027	Zirconia
E3	Dielectric installation quality assurance protocol	Method for verifying correct installation of dielectric building materials using non-contact impedance measurement and thermal imaging	Q2 2027	Zirconia

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SECTION 3: PATENT FILING STRATEGY & TIMELINE

Filing Tiers

TIER 1 — IMMEDIATE FILING (Q3 2026 — Q4 2026): 8 inventions Provisional patent applications, 12-month window to convert to PCT/non-provisional. - A1: Fluoride-free MXene synthesis (blocking patent for CSM's entire MXene platform) - A2: Discontinuous tile MXene EMI shield (core to all products using MXene coating) - B1: Dielectric Citadel 7-layer panel (flagship product architecture) - B7: Neural-Grip MRE steering wheel (most commercially developed, fastest path to revenue) - B8: Cervical-Guard headrest (second most developed, strong clinical data) - E1: 5-Phase Phantom test protocol (defensive — prevent competitors from claiming the testing methodology) - C1: LandOLil GIC risk assessment system (software patent, easier to obtain) - D4: Discontinuous MXene tile coating process (manufacturing IP to complement A2 composition patent)

Estimated Tier 1 Filing Cost: $120K–$180K (8 provisional applications at $15K–$22K each, including prior art search and drafting)

TIER 2 — Q1 2027 — Q2 2027: 14 inventions - A3, A4, B2, B3, B5, B6, B9–B16, B18, D1, D3

Estimated Tier 2 Filing Cost: $210K–$310K

TIER 3 — Q3 2027 — Q4 2027: 12 inventions - A5, B4, B17, B19, B20, C2–C5, D2, D5, D6, E2, E3

Estimated Tier 3 Filing Cost: $180K–$265K

TIER 4 — 2028+: 8 inventions (as technology matures) - Phantom MK-2 improvements, Ascension drone swarm, Charlemagne Fleet II, additional embodiments

Total 5-Year Patent Filing Budget: $750K–$1.1M (estimated, depends on international filing strategy — PCT vs. direct national filing)

Geographic Filing Strategy

Market	Patent Protection Rationale	Filing Approach
United States	Primary market; all inventions file here first (provisional)	File all 42 inventions as provisionals, convert highest-value to non-provisionals (target 25 non-provisionals over 5 years)
European Union (EPO)	Second market; Charlemagne Fleet, Protonic Network, Aegis Home (EU Green Deal alignment)	PCT → EPO regional phase for top 15 inventions
Japan	Advanced materials market; Phantom robot, piezoelectric elements, ceramic bearings	PCT → Japan national phase for top 10 inventions
China	Manufacturing base risk (copycat risk high); enforcement challenging but filing creates licensing leverage	PCT → China national phase for top 5 inventions (composition of matter only — enforcement practical via Customs recordation)
South Korea	Shipbuilding hub; Charlemagne Fleet retrofit market	PCT → Korea national phase for vessel hardening inventions
Singapore	APAC maritime hub; Phase III manufacturing base	PCT → Singapore for top 8 inventions
International (PCT)	All inventions; preserves option to enter national phase in 157 countries	File PCT for all Tier 1 and Tier 2 inventions (22 total)

SECTION 4: TRADE SECRET STRATEGY

Trade Secret vs. Patent Decision Framework

Factor	Patent	Trade Secret
Reverse-engineerable?	YES → Patent	NO → Trade Secret
Detectable in final product?	YES → Patent	NO → Trade Secret
Long commercial life? (>20 years)	Trade Secret (patent expires after 20)	Trade Secret
Enforceable? (can detect infringement?)	YES → Patent	Not applicable
Employee mobility risk?	Patent (public, employees can’t take it)	Trade Secret (risk of disclosure by departing employees)
Patent landscape crowded? (narrow claims likely)	Trade Secret (narrow patent has limited value)	Trade Secret

CSM Trade Secret Inventory

The following are PROTECTED AS TRADE SECRETS and will NOT be patented:

Trade Secret	Rationale for Non-Patenting
MXene etching process parameters (electrolyte composition, current density, temperature profile, duration)	Not detectable from final MXene product; process is conducted in-house; 20-year patent would expire before peak market; protects CSM's cost advantage indefinitely
Flash sintering parameters for ZrB_2-SiC (current waveform, ramp rate, pressure profile)	Not detectable from sintered ceramic; manufacturing process IP; indefinite protection
BFRP/Elium pultrusion die geometry and temperature profile	Not detectable from pultruded product; process know-how; indefinite protection
$KNbO_3$-$BaTiO_3$ dopant levels and calcination schedule	Detectable only via destructive analysis (XRD, ICP-MS) which is expensive and uncertain; competitor would need to independently discover optimal composition
YInMn Blue calcination temperature profile and atmosphere	Not detectable from finished pigment; process IP; indefinite protection
LandOLil source code (specific algorithms for GIC risk scoring, premium credit calculation)	Protected by copyright (automatic) + trade secret; compiled code in app store is not reverse-engineerable to source
CSM supplier list and pricing agreements	Commercially sensitive; indefinite protection via confidentiality agreements
Customer/partner pipeline and negotiation positions	Commercially sensitive; indefinite protection
Phantom MK-1 actuator calibration data and motion control algorithms	Embedded firmware, not practically extractable; indefinite protection + copyright
H-FET fabrication yield optimization parameters	Process know-how; indefinite protection

Trade Secret Protection Measures

Measure	Implementation	Timeline	Cost
Employee Confidentiality Agreements	All employees, contractors, and consultants sign comprehensive NDAs with invention assignment clauses	In place for all current personnel; ongoing for new hires	\$5K/year (legal review)
Visitor Access Control	CSMFAB facility: badge access, visitor escorts, no photography in MXene/BFRP production areas	Q4 2026 (Phase I facility)	\$85K (access control system)
Digital Security	Encrypted servers, MFA for all systems, no portable storage devices in production areas, air-gapped network for trade secret documentation	Q3 2026	\$45K setup + \$15K/year
Document Marking	All trade secret documents watermarked “CSM TRADE SECRET — DO NOT COPY” with digital rights management	Q3 2026	\$12K (DRM software)
Need-to-Know Access	Production process documentation segmented — no single employee has access to complete process; operators know their station parameters only	Q4 2026 (Phase I implementation)	\$8K (document segmentation)
Exit Interviews	Departing employees reminded of ongoing confidentiality obligations; return all CSM materials; 12-month non-compete for	Ongoing	\$3K/year

key technical personnel

SECTION 5: DEFENSIVE PUBLISHING STRATEGY

Purpose

Prevent competitors from patenting obvious extensions of CSM's technology by publishing them into the public domain. Defensive publications create prior art that blocks future patent claims by others.

CSM Defensive Publication Candidates

Topic	Rationale	Publication Venue	Timeline
General concept of MXene for residential EMI shielding	Prevent a competitor from patenting the broad concept and blocking CSM	IP.com (defensive publication database), Research Disclosure	Q3 2026
Use of BFRP rebar for GIC reduction in foundations	Protect the concept CSM is commercializing without the cost of patent prosecution	IP.com, open-access journal (Materials, MDPI)	Q4 2026
PMMA POF for in-building communication during EMI events	General concept — not a CSM competitive advantage (off-the-shelf technology)	IP.com	Q1 2027
Geopolymer concrete for EMI-resistant structures	Prevent construction material competitors from patenting the application	ACI Materials Journal (open access)	Q1 2027
General architecture of a GIC-hardened residential building	"Prior art" for the broad building-system concept	IP.com, open-access architecture journal	Q2 2027
$LiFePO_4$ battery selection for GIC-hardened applications	Battery chemistry selection is public knowledge; publishing prevents anyone from patenting the specific application	Journal of Energy Storage (open access)	Q3 2027
Use of Si_3N_4 bearings for dielectric motor applications	General concept — prevent bearing manufacturer from asserting patent	IP.com	Q4 2026
Concept of capacitive Schumann Resonance coupling via vehicle touchpoints	Protect the general concept while patenting the specific implementation (Neural-Grip)	IP.com	Q1 2027

Annual Defensive Publishing Budget: $12K–$18K

SECTION 6: LICENSING STRATEGY

Inbound Licensing (CSM as Licensee)

Technology	Licensor	Royalty Estimate	Rationale
MXene composition-of-matter ($Ti_3C_2T_x$)	Drexel University	2–4% net sales on MXene-containing products	Required if CSM's fluoride-free MXene is found to be equivalent to Drexel's patented composition. Negotiate 0% royalty for humanitarian/public-benefit applications
			OSU holds composition

YInMn Blue pigment	Oregon State University	1–2% net sales on YInMn Blue-containing products	patent on YInMn Blue; CSM’s $CsPbBr_3$-enhanced variant may be a derivative work requiring license
LORD MRF-140CG	Parker Hannifin (LORD)	N/A (purchased material)	No IP license needed; CSM purchases MR fluid as a component. CSM’s application (Neural-Grip, Fluid-Damp) is independently patentable
Nafion 115	Chemours	N/A (purchased material)	No IP license needed for purchase. If CSM modifies Nafion (e.g., surface treatment for MXene gate adhesion), the modified membrane is a derivative work — may be independently patentable

Outbound Licensing (CSM as Licensor)

Phase 1 (2028–2030): AEGIS CERTIFIED Quality Mark - License the AEGIS CERTIFIED mark to qualified builders, automotive OEMs, and vessel operators - Annual certification fee: $5,000–$50,000 depending on entity size - Requires compliance with CSM installation specifications, use of CSM-approved materials, and passing CSM-administered post-installation testing - Revenue: $2–5M/year at scale (500+ certified entities) - IP protected by trademark registration (AEGIS CERTIFIED™, DIELECTRIC CITADEL™, and the turquoise-zirconia shield emblem)

Phase 2 (2030+): Technology Licensing - License CSM’s MXene spray-coating process to qualified coating applicators (non-competitive markets) - License Phantom actuator technology for non-competing applications (prosthetics, industrial automation) - License Protonic H-FET for academic/research use (royalty-free for non-commercial) - Revenue: $8–15M/year at scale

Phase 3 (2032+): Cross-Licensing with Major Corporations - Automotive OEMs: Cross-license Aegis Embark patents for access to OEM distribution channels - Construction material companies: Cross-license Aegis Home patents for manufacturing scale - Defense contractors: Cross-license Phantom/Hardening patents for government contract access - Insurance carriers: Cross-license risk assessment algorithms for actuarial data access

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SECTION 7: FREEDOM-TO-OPERATE SUMMARY

Product Line	FTO Rating	Key IP Risks	Mitigation	Clearance Target
MXene Shielding (all products)	AMBER	Drexel $Ti_3C_2T_x$ composition patent (US9,193,595), EMI shielding article patent (US11,021,374)	License negotiation + invent-around (fluoride-free alternative)	Q1 2027
Aegis-C Composite Panel	GREEN	No blocking patents identified	File CSM patent (B1) before public disclosure	Q4 2026
Aegis Home (all 18 products)	GREEN	No blocking patents for dielectric residential building system	File CSM patents + defensive publications	Q3 2027
Phantom MK-1 Robot	GREEN	Boston Dynamics legged locomotion patents (narrow — hydraulic actuation, not piezoelectric)	CSM uses piezoelectric actuation — clearly outside Boston Dynamics claims	Q1 2027
		No prior art for EM-	File CSM patent (B3)	

Charlemagne Fleet	GREEN	hardened cruise vessel	before first public vessel retrofit disclosure	Q2 2027
Protonic Communications	GREEN	No blocking patents — pioneering field	File aggressively to establish patent thicket before academic groups enter the field	Q3 2027
Aegis Embark (all 8 products)	GREEN	LORD MRF-140CG material patent (expired or nonexistent — material is sold as commodity)	No IP risk from purchased components; CSM applications are novel	Q1 2027

Overall FTO Assessment: 6 GREEN, 1 AMBER (MXene). The AMBER risk on MXene is manageable via license negotiation, invent-around strategy, and social-impact licensing for public-benefit applications.

SECTION 8: IP BUDGET — 5-YEAR PROJECTION

Category	2026 (Q3-Q4)	2027	2028	2029	2030	5-Year Total
Patent Filing (Tier 1–4)	$180K	$310K	$265K	$120K	$80K	$955K
Patent Prosecution (office actions, responses)	$0	$45K	$120K	$180K	$220K	$565K
PCT International Filing + National Phase	$0	$85K	$180K	$220K	$250K	$735K
Prior Art Searches (per invention)	$45K	$65K	$50K	$30K	$20K	$210K
IP Legal Counsel (retainer + hourly)	$90K	$180K	$220K	$250K	$280K	$1.02M
Defensive Publishing	$8K	$18K	$15K	$12K	$10K	$63K
Trademark Registration (AEGIS CERTIFIED, DIELECTRIC CITADEL, CSM marks)	$15K	$25K	$15K	$10K	$10K	$75K
FTO Opinions (formal legal opinions for investors)	$0	$85K	$45K	$0	$0	$130K
Inbound License Fees (Drexel, OSU)	$0	$50K	$120K	$200K	$350K	$720K
Trade Secret Protection (systems, training)	$65K	$45K	$30K	$30K	$30K	$200K

ANNUAL TOTAL	**$403K**	**$908K**	**$1.06M**	**$1.05M**	**$1.25M**	**$4.67M**

"Intellectual property is the Dielectric Citadel of the balance sheet — invisible to the naked eye, intangible in the hand, but the only thing standing between the company and the competitive storm that will inevitably follow when the world realizes Carrington hardening is not a luxury but an existential necessity. File early. File aggressively. Trade-secret the processes. Defensive-publish the obvious. License the brands. And never, ever disclose the MXene etching parameters." — Director Kairos Steele, CITADEL
